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# P2 — Forces

OCR GCSE Physics A (J249) — Gateway Science · Paper 2

## P2.1 Scalars and Vectors

A **scalar** has size (magnitude) only. A **vector** has both magnitude and direction. Arrows on diagrams represent vectors — length shows size, direction of the arrow shows direction of the quantity.

| Scalar quantities | Vector quantities |
|---|---|
| Distance, speed, mass, time, energy, temperature | Displacement, velocity, acceleration, force, momentum, weight |

Notice the pairs: **distance / displacement** and **speed / velocity**. Walking 5 m north then 5 m south gives a distance of 10 m, but the displacement (straight-line from start to finish) is 0 m. The direction matters.

## P2.2 Contact and Non-contact Forces

A **force** is a push or a pull on an object, measured in **newtons (N)**. Forces are vectors. They can be **contact** (objects must touch) or **non-contact** (forces act at a distance through fields).

| Contact forces | Non-contact forces |
|---|---|
| Friction, air resistance, drag, tension, normal contact force, upthrust, applied/push force | Gravitational, electrostatic, magnetic |

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## Weight vs Mass

**Mass** is the amount of matter in an object — measured in kg; unchanged anywhere in the universe.
**Weight** is the *force* of gravity on an object — measured in N; changes depending on gravitational field strength.

$$\mathbf{W = m \times g} \quad (\text{N}) = (\text{kg}) \times (\text{N/kg})$$

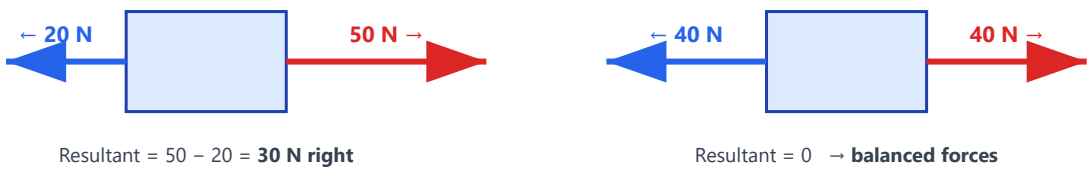
On Earth $g \approx 9.8$ N/kg (often rounded to 10). On the Moon $g \approx 1.6$ N/kg.

**Worked Example:** An astronaut has mass 80 kg. Their weight on Earth is 80 × 9.8 = **784 N**. Their weight on the Moon is 80 × 1.6 = **128 N**. Mass is unchanged at 80 kg everywhere.

## Resultant Force

When several forces act on an object, the **resultant force** is the single force that would have the same effect as all of them combined. If forces are along a straight line, just add or subtract them; forces in the same direction add, opposing forces subtract.

**Adding forces along a line**




*Balanced forces: object is either stationary or moving at constant velocity.*

## Perpendicular Forces — Scale Diagrams

When two forces act at right angles to each other, draw a scale diagram (tip-to-tail) and use a ruler + protractor to measure the resultant. You can also use Pythagoras: the magnitude of the resultant of two perpendicular forces of size a and b is $\sqrt{(a^2 + b^2)}$.

# P2.3 Work Done

**Work done** is the energy transferred when a force moves an object. 1 joule = 1 newton × 1 metre. So doing 1 J of work is the same as transferring 1 J of energy.

$$\mathbf{W = F \times d} \quad (\mathrm{J}) = (\mathrm{N}) \times (\mathrm{m})$$

**Worked Example:** A horse pulls a cart with a force of 400 N for 25 m. Work done = 400 × 25 = **10 000 J**. This energy is transferred mostly to kinetic energy of the cart, plus some to heat via friction.

Work done against **friction** heats up the surfaces in contact. That's why rubbing your hands warms them and why car brakes get hot during braking.

# P2.4 Motion — Describing How Things Move

## Speed, Velocity, Acceleration

$$\textbf{speed = distance ÷ time} \quad (\mathrm{m/s}) = (\mathrm{m}) / (\mathrm{s})$$

$$\textbf{acceleration = change in velocity ÷ time} \quad a = (v - u) / t \quad (\mathrm{m/s^2})$$

| Typical speeds (m/s) | Value |
|---|---|
| Walking | 1.5 |
| Running | 3 |
| Cycling | 6 |
| Car (urban) | 13 (≈ 30 mph) |

| | |
|---|---|
| Train | 55 |
| Sound in air | 330 |
| Wind (breeze → storm) | 5 → 30 |

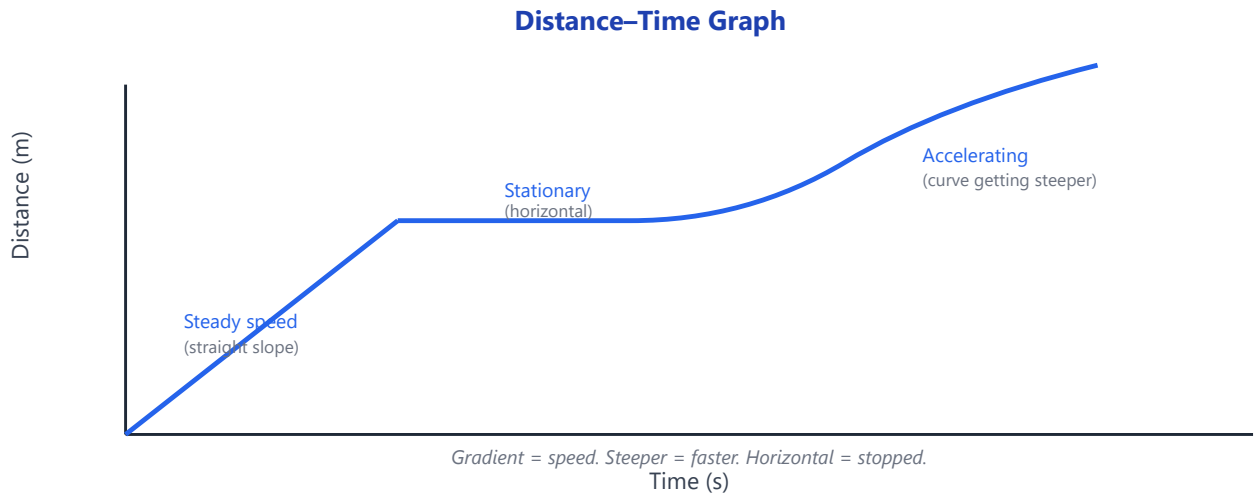
## Uniform Acceleration Equation

$$v^2 = u^2 + 2 \times a \times s$$

v = final velocity (m/s), u = initial velocity (m/s), a = acceleration (m/s²), s = distance (m)

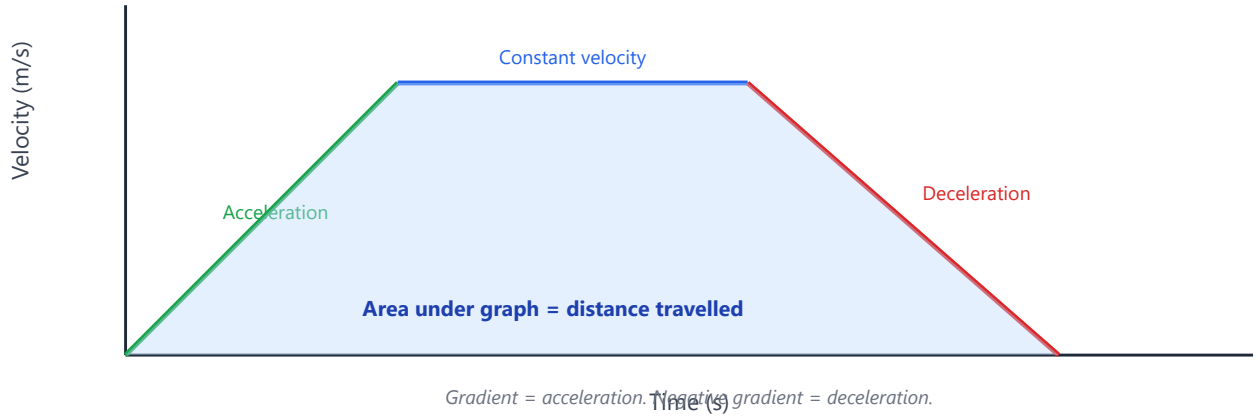
**Worked Example:** A car accelerates from rest at 2 m/s² over a distance of 50 m. Final velocity? $v^2 = u^2 + 2as = 0 + 2 \times 2 \times 50 = 200$. So $v = \sqrt{200}$ = **14.1 m/s**.

## Distance-Time Graphs




## Velocity-Time Graphs

**Velocity–Time Graph**




*Gradient = acceleration. Negative gradient = deceleration.*

**Exam Tip:** Do not confuse distance-time and velocity-time graphs.
• Distance-time: gradient = speed, horizontal line = stationary.
• Velocity-time: gradient = acceleration, horizontal line = constant velocity, area under graph = distance.

# P2.5 Newton's Laws of Motion

## Newton's First Law — Inertia

> An object continues at rest or at constant velocity unless acted on by a resultant force. Another way: if forces are balanced, velocity doesn't change.

This is why seatbelts matter: during a sudden stop, your body continues moving forward until an external force (the belt or the windshield) stops it. The tendency to keep doing what you were doing is called **inertia**.

## Newton's Second Law

$$\mathbf{F = m \times a} \quad (\mathrm{N}) = (\mathrm{kg}) \times (\mathrm{m/s^2})$$

A resultant force on a mass causes an acceleration in the same direction.

**Worked Example:** A car of mass 1200 kg accelerates at 3 $m/s^2$. Resultant force needed? F = ma = 1200 × 3 = **3600 N**.
If the same 3600 N acted on a lighter car of mass 800 kg, acceleration would be 3600/800 = **4.5 $m/s^2$**. Lighter mass → bigger acceleration for the same force.

**Inertial mass** is defined by F = ma: it is the property that resists a change in motion. A heavier mass needs a bigger force to produce the same acceleration.

### Newton's Third Law

When two objects interact, the forces they exert on each other are **equal in size and opposite in direction**. Every action has an equal and opposite reaction.

Examples:

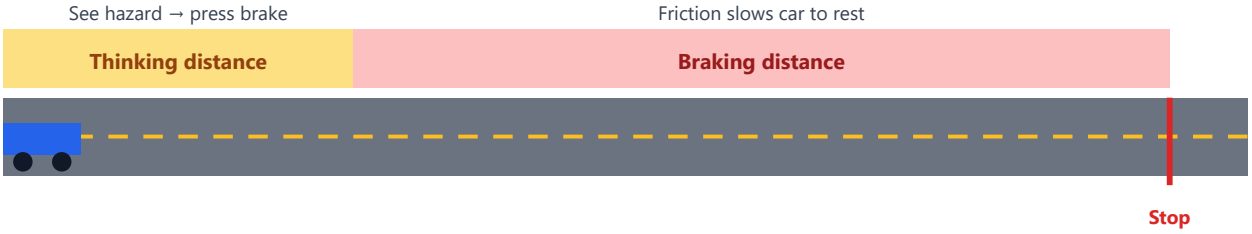
- You push down on a chair; the chair pushes up on you with the same force (normal contact force).
- A rocket pushes exhaust gas downwards; the gas pushes the rocket upwards with equal force.
- Earth pulls you down with your weight; you pull Earth up with the same force (Earth is far too massive to notice).

**Exam Tip:** The action and reaction act on *different* objects. This is easy to confuse with balanced forces (which act on the *same* object). A ball on a table: the weight (Earth pulls ball) and normal contact force (table pushes ball) are balanced forces on the ball — NOT a Newton's-third-law pair.

## P2.6 Stopping Distances

**Stopping distance** = **thinking distance** + **braking distance**.
Thinking distance = distance travelled while the driver reacts (during reaction time).
Braking distance = distance travelled while the brakes slow the car to rest.

**Stopping Distance = Thinking + Braking**




## Factors Affecting Thinking Distance

Thinking distance depends on the driver's **reaction time** (≈ 0.2 – 0.9 s) and the vehicle's **speed**:

- Alcohol, drugs, tiredness — slow reactions, longer thinking distance.
- Using a phone / distractions — same effect.
- Higher speed — more distance covered in the same reaction time.

## Factors Affecting Braking Distance

- **Speed** — braking distance is proportional to $\mathbf{v^2}$. Doubling speed quadruples braking distance.
- **Road surface** — wet, icy, oily or gravelled roads reduce friction, increase braking distance.
- **Tyre / brake condition** — worn tyres or worn brake pads are less effective.
- **Mass of vehicle** — heavier vehicles (loaded lorry) need more force to decelerate, so longer braking distance.

| Speed (mph) | Thinking (m) | Braking (m) | Total stopping (m) |
|---|---|---|---|
| 20 | 6 | 6 | 12 |
| 30 | 9 | 14 | 23 |
| 40 | 12 | 24 | 36 |
| 50 | 15 | 38 | 53 |
| 60 | 18 | 55 | 73 |
| 70 | 21 | 75 | 96 |

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### Hard Braking — Dangers

Bringing a fast-moving car to rest requires a **large deceleration**, which needs a **large force** (F = ma).

- The brakes do a huge amount of work in a short distance — they heat up dramatically.
- If brakes overheat, the friction pads lose grip (**brake fade**).
- Large deceleration can cause loss of control, especially if wheels skid on a low-friction surface.

## P2.7 Momentum

**Momentum (p)** is a measure of how much motion an object has. It is mass × velocity. Momentum is a **vector**. Units: kg m/s.

$$\mathbf{p} = \mathbf{m} \times \mathbf{v} \quad (\text{kg m/s}) = (\text{kg}) \times (\text{m/s})$$

### Conservation of Momentum

In a closed system (no external forces) the total momentum before an event equals the total momentum after. This is called **conservation of momentum**.

### Force and Momentum Change

A force changes momentum. The **larger the force** or **longer the time** the force acts, the greater the change in momentum.

$$F = (mv - mu) / t \quad \text{(alternative form of } F = ma\text{)}$$

where (mv − mu) is the change in momentum.

### Why Safety Features Work

Crumple zones, airbags, seatbelts and helmets all work on the same principle: they **increase the time** over which a person's momentum is reduced to zero, which **reduces the force** on them.

| Safety feature | How it works |
|---|---|
| **Seatbelt** | Stretches slightly — increases stopping time, reduces force on chest. |
| **Airbag** | Inflates between driver and dashboard — longer deceleration, lower peak force. |
| **Crumple zone** | Front/rear of car crushes — absorbs energy, extends collision time. |
| **Cycle helmet** | Foam liner crushes on impact — extends time, spreads force over head. |
| **Crash mats** | Foam compresses — increases landing time, reduces impact force. |

**Worked Example:** A 70 kg driver in a crash comes to rest from 15 m/s in 0.1 s (no seatbelt) or 0.5 s (with seatbelt + airbag).
Change in momentum = 70 × 15 = 1050 kg m/s.
• No restraint: F = 1050 / 0.1 = **10 500 N**.
• With restraint: F = 1050 / 0.5 = **2 100 N**.
Five times longer stopping time → five times less force. That can be the difference between life and death.

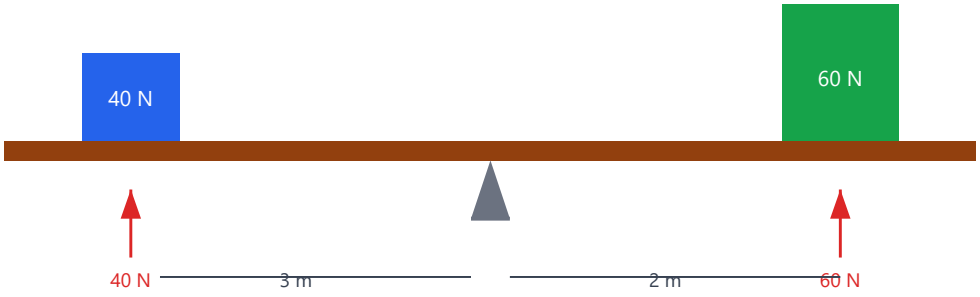
## P2.8 Moments — Turning Effects

The **moment** of a force is its turning effect about a pivot. A longer lever gives a bigger turning effect for the same force. Unit: N m.

$$\textbf{Moment} = \mathbf{F} \times \mathbf{d} \quad (\text{N m}) = (\text{N}) \times (\text{m})$$

d is the *perpendicular* distance from pivot to line of action.

**Principle of Moments — Balanced Seesaw**




**40 × 3 = 120 N m = 60 × 2 = 120 N m → balanced**

## Principle of Moments

For a balanced object: **total clockwise moment = total anticlockwise moment**.

> **Worked Example:** A child weighing 200 N sits 2.0 m from a pivot. How far from the pivot must a 160 N child sit on the other side for balance?
> Clockwise = anticlockwise → 200 × 2.0 = 160 × d → d = 400 / 160 = **2.5 m**.

## Levers and Gears

Both are **force multipliers**: they let a small input force produce a large output force, at the cost of having to move a larger distance.

- **Levers** (spanners, crowbars, wheelbarrows): a long handle with a short load distance multiplies force.
- **Gears** interlocking wheels: a small driver gear turning a large driven gear transmits a larger turning force (but at slower speed).

# Quick Summary — Key Equations for P2

| Quantity | Equation | Units |
|---|---|---|

| | | |
|---|---|---|
| Weight | $W = m\,g$ | N |
| Work done | $W = F\,d$ | J |
| Speed | $v = d / t$ | m/s |
| Acceleration | $a = (v - u) / t$ | $m/s^2$ |
| Newton's 2nd law | $F = m\,a$ | N |
| Uniform accel | $v^2 = u^2 + 2\,a\,s$ | — |
| Momentum | $p = m\,v$ | kg m/s |
| Force and momentum | $F = (mv - mu)/t$ | N |
| Moment | $M = F\,d$ | N m |